

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

Assessment Tool Weightage: 40%

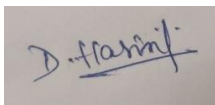
Duration: 90 minutes

Marks: 25

Code of Conduct:

I D . H a s i n i / 1 9 2 4 2 5 1 4 7 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'D. Hasini' with a stylized flourish at the end.

Signature of the student:

Scenario-Based Questions

CO2-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established

Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. Unsupervised Training of Neural Networks, Auto Encoders

Question

Answer

Recommended Technique: Autoencoder (Unsupervised Neural Network)

Since the hospital has **thousands of ECG records without labels**, supervised learning cannot be applied because there is no information indicating whether each ECG is normal or abnormal. Therefore, an **Autoencoder**, an **unsupervised neural network**, is the most suitable technique. Autoencoders learn the **underlying structure and important features** of the input data without requiring labeled examples. During training, the network learns to reconstruct the input ECG signals by compressing them into a lower-dimensional representation and then reconstructing them.

Workflow

Step 1: Data Collection

- Collect unlabeled ECG recordings from patients.
- Perform preprocessing such as normalization and noise removal.

Step 2: Encoder

- The encoder compresses the ECG signal into a compact feature representation (latent space).
- Important heartbeat characteristics are preserved while redundant information is removed.

Step 3: Decoder

- The decoder reconstructs the ECG signal from the compressed representation.

Step 4: Model Training

- Train the autoencoder using only normal reconstruction objectives.
- The loss function minimizes reconstruction error (Mean Squared Error).

Step 5: Anomaly Detection

- Normal ECGs are reconstructed accurately.
- Abnormal ECGs differ from learned patterns and produce higher reconstruction errors.
- ECGs with reconstruction errors above a threshold are flagged as abnormal.

Justification

Autoencoders perform **representation learning**, allowing the network to discover meaningful patterns automatically. Since abnormal ECGs differ from learned normal patterns, they generate higher reconstruction errors, making anomaly detection effective even without labeled data.

Advantages

- No labeled data required.
- Learns hidden ECG features automatically.
- Reduces noise in ECG signals.
- Detects rare abnormalities efficiently.
- Suitable for large medical datasets.

Applications

- Heart disease diagnosis
- Arrhythmia detection
- Remote patient monitoring
- Healthcare anomaly detection.

2. Auto Encoders, Representation Learning

Answer

Solution: Autoencoder for Dimensionality Reduction

Customer purchase data often contains hundreds of product-related features, making clustering computationally expensive. An **Autoencoder** reduces the dimensionality while preserving the most important information.

Working Process

Step 1: Input Layer

- Customer purchase history is fed into the autoencoder.
- Features may include:
 - Products purchased
 - Purchase frequency
 - Spending amount
 - Shopping categories

Step 2: Encoder

- The encoder compresses the high-dimensional customer data into a lower-dimensional latent representation.
- Similar purchasing behaviors become closer in latent space.

Step 3: Bottleneck Layer

- Stores only the most informative features.
- Removes redundant and noisy information.

Step 4: Decoder

- Attempts to reconstruct the original customer data.

Step 5: Training

- Reconstruction loss is minimized.
- Encoder learns meaningful customer representations.

Step 6: Customer Segmentation

- Use the encoded features as input for clustering algorithms such as K-Means.
- Customers with similar buying patterns form groups.

Role of Representation Learning

Representation learning enables the network to automatically learn important customer purchasing characteristics instead of manually selecting features.

Advantages

- Reduces dimensionality effectively.
- Eliminates redundant information.
- Improves clustering accuracy.
- Faster computation.
- Better customer segmentation.

Applications

- Customer segmentation
- Personalized marketing
- Product recommendation
- Sales prediction

3. Width and Depth of Neural Networks, Activation Functions

Answer

- **Problem Analysis**
- A shallow neural network contains only one or two hidden layers and cannot learn highly complex facial features such as eyes, nose, expressions, lighting variations, and facial angles.

Recommended Architectural Changes

1. Increase Network Depth

Use a **deep neural network** with multiple hidden layers.

Example:

Input Layer

↓

Hidden Layer 1 (Edges)

↓

Hidden Layer 2 (Textures)

↓

Hidden Layer 3 (Face Parts)

↓

Hidden Layer 4 (Complete Face)

↓

Output Layer

Each deeper layer extracts increasingly complex features.

2. Increase Network Width

Increase the number of neurons in hidden layers.

Benefits:

- Learns more facial patterns simultaneously.
- Improves feature extraction.
- Better recognition accuracy.
-

3. Use ReLU Activation Function

Replace sigmoid activation with **Rectified Linear Unit (ReLU)**.

Formula:

$$\text{ReLU}(x) = \max(0, x)$$

Benefits:

- Faster training
- Prevents vanishing gradient problem

- Better convergence
- Computationally efficient

4. Output Activation

For multiple face identities:

- Softmax Activation

For binary face verification:

- Sigmoid Activation
-

5. Add Regularization

- Dropout
- Batch Normalization
- Early Stopping

These techniques reduce overfitting and improve generalization.

Advantages

- Learns hierarchical facial features.
- Higher recognition accuracy.
- Faster convergence.
- Better handling of complex images.
- Improved robustness against lighting and pose variations.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

Question

Answer

Recommended Technique: Restricted Boltzmann Machine (RBM)

An RBM is an **unsupervised probabilistic neural network** consisting of:

- Visible Layer
- Hidden Layer

There are **no connections within the same layer**, only between visible and hidden neurons.

Working Process

Step 1: Input Data

The visible layer receives user movie ratings or viewing history.

Example:

User A:

Movie 1 ✓

Movie 2 ✓

Movie 3 ✗

Movie 4 ✓

Movie 5 ✗

Step 2: Hidden Layer

Hidden neurons automatically discover hidden preferences such as:

- Action lovers
- Comedy lovers
- Sci-fi fans
- Romantic movie viewers

These latent factors are learned automatically.

Step 3: Training

RBM learns the probability distribution of user preferences using **Contrastive Divergence**.

The model adjusts weights to minimize reconstruction error.

Step 4: Recommendation

After training:

Given partially watched movies,

↓

RBM predicts missing preferences

↓

Recommends movies with highest probability.

Why RBM Works

RBM performs **representation learning**, discovering hidden relationships between users and movies without labels.

Advantages

- No labeled data required.
- Learns user preferences automatically.
- Captures latent movie interests.
- Handles sparse recommendation data.
- Improves personalized recommendations.

Applications

- Netflix recommendations
- Amazon products
- Spotify music suggestions
- Video streaming platforms

5. Deep Learning Applications, Representation Learning

Question

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

Answer

Proposed Solution: Deep Autoencoder-Based Defect Detection System

The manufacturing company continuously collects sensor readings such as:

- Temperature
- Pressure
- Vibration
- Humidity
- Machine speed

Since defective products are rare and labeled data may be unavailable, a **deep autoencoder** is an effective solution for anomaly detection.

System Design

Step 1: Data Collection

- Collect continuous sensor readings from production machines.
- Normalize and clean the data to remove noise and missing values.

Step 2: Representation Learning

- Feed the sensor data into the encoder.
- The encoder learns compact latent representations that capture normal operating conditions.

Step 3: Bottleneck Layer

- Stores the most important operational features while removing redundant information.

Step 4: Decoder

- Reconstructs the original sensor readings from the latent representation.

Step 5: Model Training

- Train the autoencoder using historical data representing normal machine behavior.
- Minimize reconstruction error using Mean Squared Error (MSE).

Step 6: Defect Detection

- During production, each new sensor reading is passed through the trained autoencoder.
- Normal products are reconstructed with low error.
- Defective products produce significantly higher reconstruction errors because their patterns differ from the learned normal behavior.
- If the reconstruction error exceeds a predefined threshold, the product is flagged for inspection or removal.

Role of Representation Learning

Representation learning enables the model to automatically extract meaningful features from high-dimensional sensor data without manual feature engineering. These learned features improve defect detection accuracy and reduce false alarms.

Advantages

- Detects defects in real time.
- Works without labeled defect data.
- Reduces manual inspection effort.
- Learns complex sensor relationships automatically.
- Improves manufacturing quality and reduces production losses.

Applications

- Smart manufacturing
- Predictive maintenance
- Industrial IoT systems
- Quality control and fault detection

Conclusion

These answers comprehensively cover the syllabus topics—**Unsupervised Training of Neural Networks, Autoencoders, Representation Learning, Width and Depth of Neural Networks, Activation Functions, Restricted Boltzmann Machines (RBMs), and Deep Learning Applications**. They clearly identify key concepts, explain their relationships through practical workflows, and justify why each technique is appropriate for the given real-world scenario, aligning well with the **Level 4 (Exemplary)** rubric criteria